



THE UNIVERSITY
OF ARIZONA

Understanding the effects of mental health well-being on the risks of developing PASC (Post-Acute Sequelae of SARS-CoV-2) infection for Arizona CoVHORT TrackPCC Participants

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RESEARCH QUESTION

Does baseline mental health and well-being influence the risk of developing PASC following COVID-19 infection?

DATA SOURCE & VARIABLES

- TrackPCC data for mental health outcomes is based on the 14-item Warwick-Edinburgh Mental Well-being Scale (WEMWBS).
- Each item is rated on a 1 ("None of the time") – 5 ("All of the time") scale, and the baseline WEMWBS score is calculated by summing the responses for all 14 items from participants who completed every question.
- The PASC status is binary, with 1 representing symptomatic and 0 representing asymptomatic between 90 and 180 days post-infection.
- Demographic factors: Age, Sex, Race, Ethnicity, Income, Education
- Final Sample Size = 222.

STATISTICAL ANALYSIS

- Logistic regression**
 - Exposure variables **WEMWBS baseline score** and **income** had small p-values of 0.000605 and 0.004486 respectively, making them statistically significant variables. Taking α (alpha) = 0.05, this indicated that having better baseline mental health well-being and a higher income level would mean lower prevalences of developing PASC.
- LASSO regression with 10-fold cross-validation**
 - Exposure variables **WEMWBS baseline score** and **income** returned non-zero coefficients of -0.011 and -0.04 respectively, making them significant predictors. This indicated that having a higher baseline mental health well-being and a higher income level predicted a lower PASC risk.

DISCUSSION

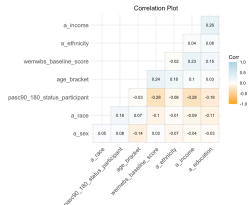
Having a better baseline mental health well-being signals lower risk of developing PASC. Findings are suggestive that incorporating extra efforts into improving mental health well-being could significantly lower the risks of an individual developing PASC. Additionally, socioeconomic factors such as income can further influence both mental health outcomes and vulnerability to PASC, highlighting the need for holistic, equity-focused prevention efforts. This emphasizes the importance of early mental health interventions and support strategies to promote mental health well-being as an integrative approach to minimize the negative health complications brought about by PASC that a patient might experience upon the onset of COVID infection.

Better baseline mental health is linked to lower likelihood of PASC.

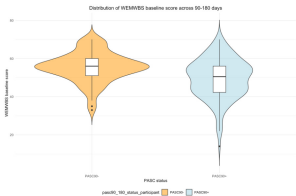
14-Item Questionnaire using the Warwick-Edinburgh Mental Well-being Scale (WEMWBS)

1. I've been feeling optimistic about the future.
2. I've been feeling useful.
3. I've been feeling relaxed.
4. I've been feeling interested in other people.
5. I've had energy to spare.
6. I've been dealing with problems well.
7. I've been thinking clearly.
8. I've been feeling good about myself.
9. I've been feeling close to other people.
10. I've been feeling confident.
11. I've been able to make up my own mind about things.
12. I've been feeling loved.
13. I've been interested in new things.
14. I've been feeling cheerful.

EXPLORATORY DATA ANALYSIS



This correlation plot shows pairwise correlations among several demographic and baseline variables, with values ranging from about -0.3 to +0.3. Most correlations are weak, and none approach moderate or strong thresholds.



The distribution of WEMWBS baseline scores across PASC outcomes, calculated using surveys between 90 – 180 days after infection suggests that PASC status influences the WEMWBS baseline score, with PASC negative participants showing a better mental health well-being as compared to PASC positive participants.

REFERENCES

1. Neo, K. S., & Falk, L. P. (2023). Understanding the effects of mental health well-being on the risks of developing PASC (Post-Acute Sequelae of SARS-CoV-2) infection for Arizona CoVHORT TrackPCC Participants. *Journal of Public Health Research and Practice*, 23(1), 1-10.

2. Neo, K. S., & Falk, L. P. (2023). Understanding the effects of mental health well-being on the risks of developing PASC (Post-Acute Sequelae of SARS-CoV-2) infection for Arizona CoVHORT TrackPCC Participants. *Journal of Public Health Research and Practice*, 23(1), 1-10.